

GenHowCC3_en: balance optimisation

This release adds tools to measure and optimise the **balance** of complete and incomplete Howell movements. Balance means how evenly every pair meets every other pair as an *indirect opponent* (same direction on the same boards) and as an *indirect partner* (opposite direction).

Highlights

- **Three buttons** replace the single "Generate" button:
 - **Generate movement** creates the movement with no balance optimisation and remembers it.
 - **Optimise balance** applies the strategy chosen in the menu, always starting from the original movement, so strategies can be compared on the same movement.
 - **Continue optimising** restarts simulated annealing from the current state. The movement changes only if the result improves, and the button does nothing if the result is already proved optimal.
- **Balance report** for every movement, with the balance percentage and a line saying which method ran, how long it took, the cost, and what is guaranteed ("optimal", "stopped by the time limit", "not guaranteed").
- **Pair/pair matrix** with the number of times each two pairs are indirect opponents, colour-coded by deviation.
- **Boards/pairs matrix** (blue = NS, pink = EW, the number is the direct opponent). Click a pair number to highlight in red the cells where that pair is the direct opponent.
- **Max optimisation time** field (default 5 s).
- **The pair with the highest number**, fixed at table 1, is always played NS in optimised movements (flipping a whole row never changes the balance).
- The user interface and all program messages are now in English. The movement list format (Table, Board, PairNS, PairEW) is unchanged.

Balancing strategies

Strategy	What it does
No optimisation	The movement as generated
NS/EW sum optimisation	Existing strategy: balances the NS/EW sums per row
Fixed pairs in NS	Existing strategy
Pairs: simulated annealing	Minimises the sum of squared deviations over all pairs of pairs
Pairs: exact search (B&B / threshold)	Incomplete Howell: threshold search that lowers the maximum deviation and then minimises the cost within that limit
Pairs: movement search	Complete Howell: runs through the movements of the generator, looks for one with perfect balance, and otherwise keeps the best found

Notes on the method

- The movement itself does not change when a match's direction is swapped, so balance can be optimised after the movement is generated.
- For a complete Howell, perfect balance is equivalent to a Hadamard matrix compatible with the movement. It is **impossible when the number of pairs is not a multiple of 4**, as long as each group of boards is played in a single direction. For those sizes the minimum cost is

- $4 \cdot C(N-1, 2)$, and the program stops when it reaches it.
- The balance percentage is $100\% \times (1 - \text{mean relative deviation})$. The mean of the ratios "opponents / ideal" is always 100% when each row has half the pairs NS and half EW, so it cannot tell good movements from bad ones.
 - For incomplete Howell movements, perfect balance for every pair of pairs is generally impossible, so the threshold search targets the smallest possible maximum deviation instead.

Results

The first movement the generator produces is not always the best one, so results depend on the movement.

Pairs	Result
8, 12	Perfect balance found at once
10, 14	Perfect impossible. Minimum cost reached (10 pairs: 84–87%, 14 pairs: 89–90%)
16	Perfect balance exists in other movements of the generator (22 of 30 sampled); the movement search finds one in a few seconds. The first movement generated is not balanceable.
18–24	Perfect not decided in a few seconds. Annealing and best-of-movements reach about 91–92%, with small differences between movements
Incomplete (10–20 pairs)	Threshold search lowers the maximum deviation in most cases, at a cost of up to a few points of average balance

Known limitations

- The exact methods do not finish at 20 pairs or more in a few seconds; there annealing is used and nothing is proved.
- The movement search is available only for complete Howell movements.
- The threshold search targets zero for every pair of pairs. A deliberate surplus of opponent rounds for pairs that never meet is not modelled.
- Splitting the boards of a group between directions at a table, which would make odd-table Howells perfect, is not supported.
- Results of the randomised methods vary from run to run.